

# Simple Network Management Protocol (SNMP) and Management Information Base II (MIB II)

Presenter: Assoc. Prof. Dr. Fatih ABUT  
Cukurova University  
Department of Computer Engineering

## History of SNMP

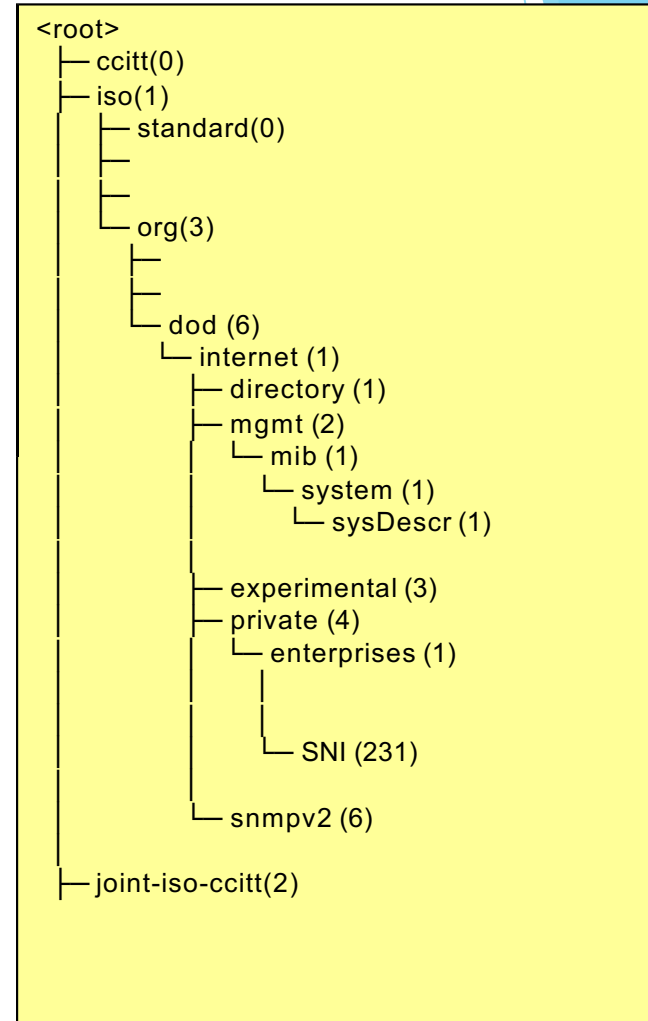
| Version                                         | SMI                                                                                                              | MIB                                                                                                                                                                                                             | Protocol                                                                                                                 |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| SGMP<br>RFC1028, 11/87                          | RFC1028, "Simple Gateway Monitoring Protocol"                                                                    |                                                                                                                                                                                                                 |                                                                                                                          |
| SNMPv1<br>RFC1065-1067, 08/88<br>RFC1098, 04/89 | RFC1065 "Structure and identification of management information for TCP/IP-based internets", (→RFC1155)          | RFC1066 "Management Information Base for network management of TCP/IP-based internets" (→RFC1156)                                                                                                               | RFC1067 "Simple Network Management Protocol" (→RFC1098)<br>RFC1098 "Simple Network Management Protocol SNMP", (→RFC1157) |
| RFC1155-1158, 05/90                             | RFC1155 (STD 17), "Structure and Identification of Management Information for TCP/IP-based Internets" (→RFC1212) | RFC1156 "Management Information Base for Network Management of TCP/IP-based internets" (→RFC1158)<br>RFC1158 "Management Information Base for Network Management of TCP/IP-based internets: MIB-II", (→RFC1213) | RFC1157 (STD 15) "A Simple Network Management Protocol (SNMP)"                                                           |
| RFC1212-1215, 03/91                             | RFC1212 (STD 16) "Concise MIB Definitions"                                                                       | RFC1213 (STD 17) "Management Information Base for Network Management of TCP/IP-based internets: MIB-II"<br>RFC1215 "A Convention for Defining Traps for use with the SNMP"                                      |                                                                                                                          |
| RFC1351-1353, 07/92                             | Secure SNMP                                                                                                      |                                                                                                                                                                                                                 |                                                                                                                          |

→ SNMP v2 vs. SNMP v3

## Object Identifier Tree (OIDtree)

Example:

sysDescr: 1.3.6.1.2.1.1.1

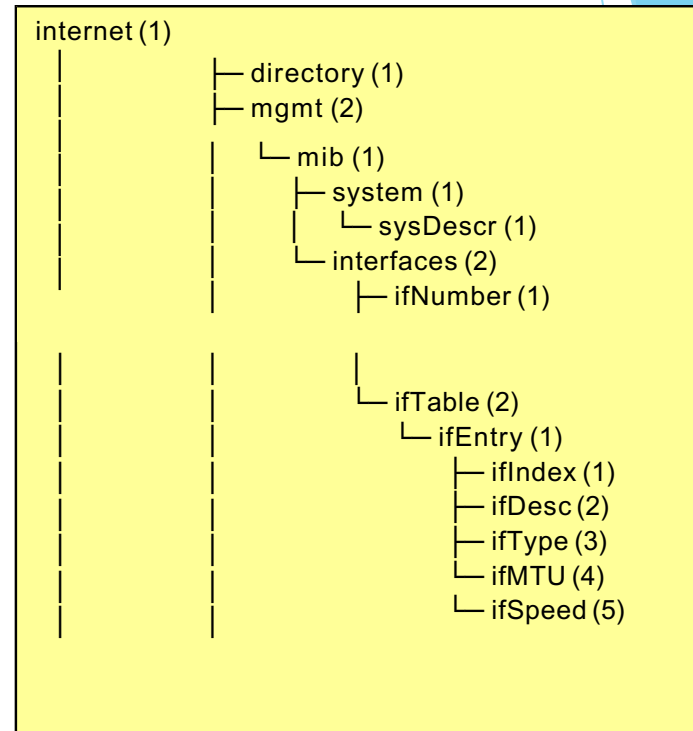


## SNMP Tables

Rationale:

- Table is built as a three-level structure in the OID tree:

- <name>Table (t) Name of the table
- <name>Entry (1) Represents a branch point for the columns of the table
- <nameAbbr><descCol> (i) columns object



## Identification of managed objects in SNMP

### Principle:

Use of the OID tree for the identification of the object instances

Schema for identifier of an SNMP object instance (SNMP variable):

`<Object Identifier>.<Suffix>`

### Case 1: Instance of a simple (non-composite) object

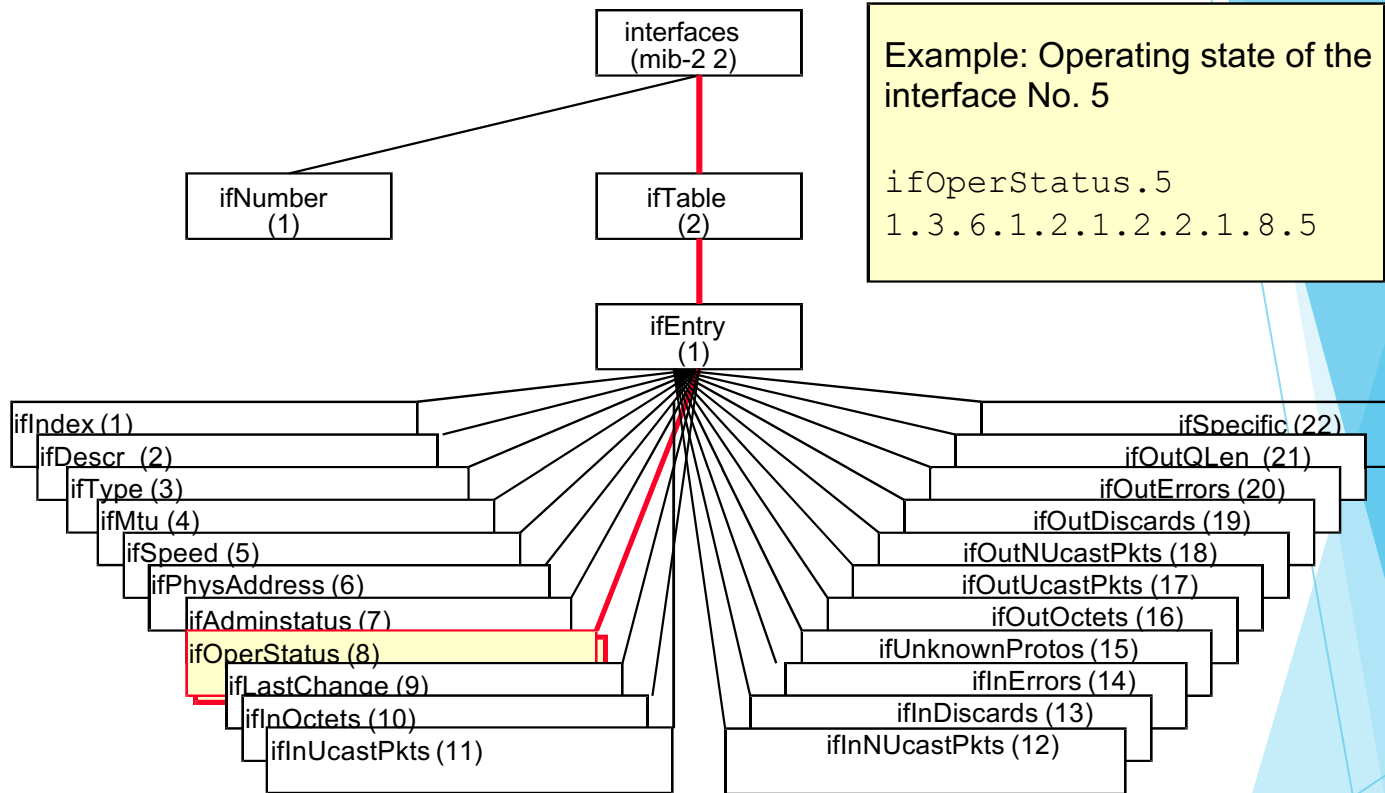
There is only one instance → suffix = 0

Example: sysDescr.0  
1.3.6.1.2.1.1.1.0

### Case 2: Instance of a column object

`<Object Identifier>` identifies column of the table

`<Suffix>` Search index in the table (specified in the MIB description)



Legend: INDEX in bold

According to Minhua Wang  
<http://www.homepages.dsu.edu/wangm/>

## Example: Status of a TCP connection

```
TcpConnEntry ::= SEQUENCE {  
  
    tcpConnState  INTEGER,  
    tcpConnLocalAddress  IpAddress,  
    tcpConnLocalPort  INTEGER (0..65535),  
    tcpConnRemAddress  IpAddress,  
    tcpConnRemPort    INTEGER (0..65535)  
}
```

OID of tcpConnState: 1.3.6.1.2.6.13.1.1

## Identification of instances of managed objects (using the status of a TCP connection as an example)

Structure of suffix:

```
INDEX { tcpConnLocalAddress, tcpConnLocalPort, tcpConnRemAddress,  
tcpConnRemPort }
```

This results in the identifier of the corresponding object instance:

```
tcpConnState.iplocal.portlocal.ipdest.portdest  
1.3.6.1.2.6.13.1.1.194.23.86.23.2045.194.23.86.15.80  
|           |           |           |           |  
|      OID      |OCTET STRING|INT|OCTET STRING|INT|
```



## Events

- Exceptional events of the resource (events) trigger a message to the manager system in the SNMP agent.
- These messages are called
  - ☐ Traps (in case of SNMPv1)
  - ☐ Notifications (in case of SNMPv2)
- An event contains an error number (and more information if needed).
- As part of SNMP, the management concept used is the so-called "trap-directed polling" model → Polling the required information after an indication by a trap).
- The traps or notifications are not confirmed in SNMP.
- Overall, the event mechanism of SNMP is judged to be very inadequate.

# Structure of Management Information (SMI)

## Structure of Management Information(SMI)

- SMI defines the ASN.1-based syntactic structure of a MIB module
- An MIB module includes:
  - Organizational information (author, ..)
  - Description of the MOs (using ASN.1 construct `OBJECT TYPE`)
  - Mount the MO in the OID tree
- Example of a MO definition using `OBJECT-TYPE`

```
ifInOctets OBJECT-TYPE
    SYNTAX Counter
    MAX-ACCESS read-only
    STATUS mandatory
    DESCRIPTION
        "The total number of octets received on the
        interface, including framing characters."
    ::= { ifEntry 10 }
```

## Simple example of general module design SNMPv2

```
COURSE-MGMT-MIB DEFINITIONS ::= BEGIN

    IMPORTS
        OBJECT-TYPE, Integer32, experimental
        FROM SNMPv2-SMI
    courseMgmt MODULE-IDENTITY
        LAST-UPDATED "050419"
        ORGANISATION „CU"
        CONTACT-INFO "Fatih Abut"

        DESCRIPTION "MIB for the management of courses"
            ::= { experimental 1 }
    participant OBJECT IDENTIFIER
        ::= { courseMgmt 1 }
    partNumber OBJECT TYPE

        SYNTAX          Integer32 (1..100)
        MAX-ACCESS       read-write
        STATUS           current
        DESCRIPTION      "Number of participants (maximal 100)"
            ::= { participant 1 }
    END --End of the module
```

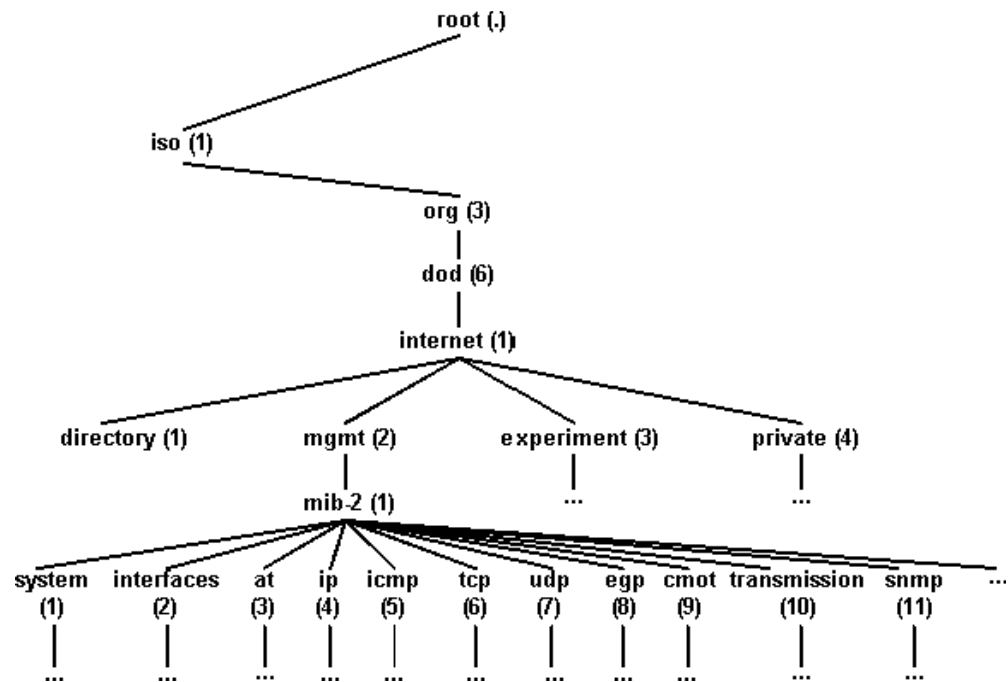
Object type macro to  
define MOs

Integration of the MO in  
OID-tree

## Standard MIB-II

- MIB-II Basic Catalog of Management Information (developed as an extension of the precursor MIB I). Management information that must be provided by each monitored system.
- Defined in RFC 1213 (+ successor)
- Related information is grouped together
- The groups can be found under the OID `mib-2 ::= {1 3 6 1 2 1}`

## Standard MIB-II in OID-Tree



## Overview of the groups of the standard MIB-II

| Group        | OID für Teilbaum | Beschreibung                                                                                                                                                                                         |
|--------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| system       | {mib-2 1}        | General information about the managed node (sysDescr, sysLocation, sysUpTime, ...)                                                                                                                   |
| interfaces   | {mib-2 2}        | <ul style="list-style-type: none"> <li>•Number of interfaces of the managed node.</li> <li>•Table containing information for each interface (ifType, ifDescr, ifSpeed, ifOutErrors, ....)</li> </ul> |
| at           | {mib-2 3}        | Address-Translation Group was part of MIB I and has been deprecated in MIB II (only for reasons of compatibility). Each protocol group will contain its own address table in the future              |
| ip           | {mib-2 4}        | Statistics counter for incoming and outgoing packets, routing table                                                                                                                                  |
| icmp         | {mib-2 5}        | Statistics counter for incoming and outgoing packets                                                                                                                                                 |
| tcp          | {mib-2 6}        | Statistics counter for incoming and outgoing packets, tcp state diagram                                                                                                                              |
| udp          | {mib-2 7}        | Statistics counter for incoming and outgoing packets                                                                                                                                                 |
| egp          | {mib-2 8}        | Statistics counter for incoming and outgoing packets                                                                                                                                                 |
| cmot         | {mib-2 9}        | Management information for CMOT (= CMIP over TCP). Status historical.                                                                                                                                |
| transmission | {mib-2 10}       | Management information about transmission protocols and network interfaces (X.25, Token Ring, FDDI, ATM, ...)                                                                                        |
| snmp         | {mib-2 11}       | Statistics counter for incoming and outgoing packets                                                                                                                                                 |

## Calculation of utilization as derived parameter

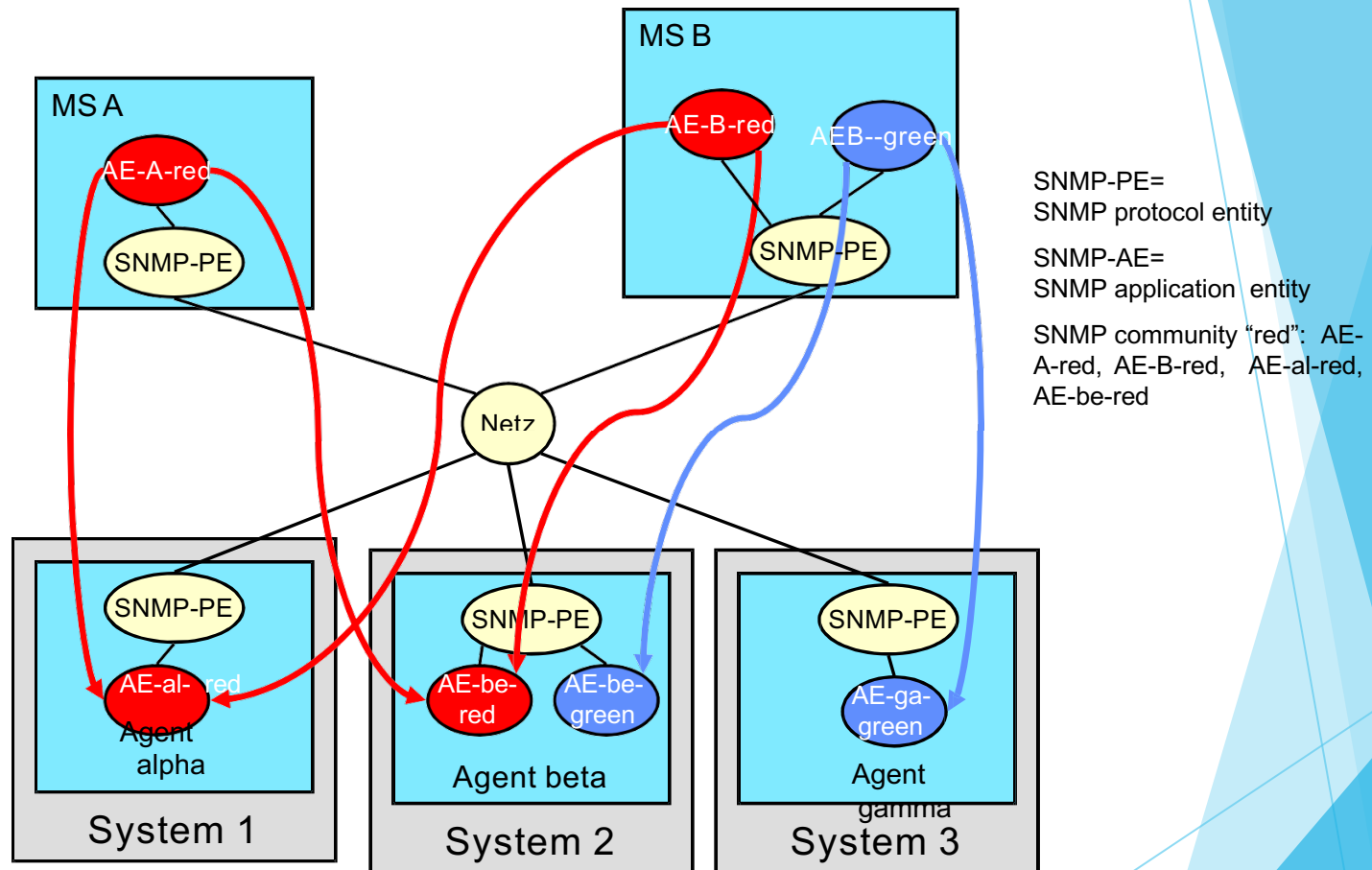
- $\text{total\_bytes} = (\text{ifInOctetsy} - \text{ifInOctetsx}) + (\text{ifOutOctetsy} - \text{ifOutOctetsx})$
- $\text{total\_bytes per sec} = \text{total\_byte} / (y - x)$
- $\text{Utilization} = (\text{total\_bytes per sec} * 8) / \text{ifSpeed}$



# Access Protocol

## The SNMP access protocol

- Objective: Simplicity (-> low development costs, easy to implement on small components, simple agent software possible)
- This objective comes at the expense of: functionality, performance, security
- The basic problems:
  - Security
  - Differentiated assignment of rights
  - Creating management views
  - Complex MS / MA relationship
- are addressed in SNMP by administrative relationships



## Community Profile

- Community

An SNMP community is a relationship between an SNMP agent and a set of SNMP managers.

Different agents can use the same community name.

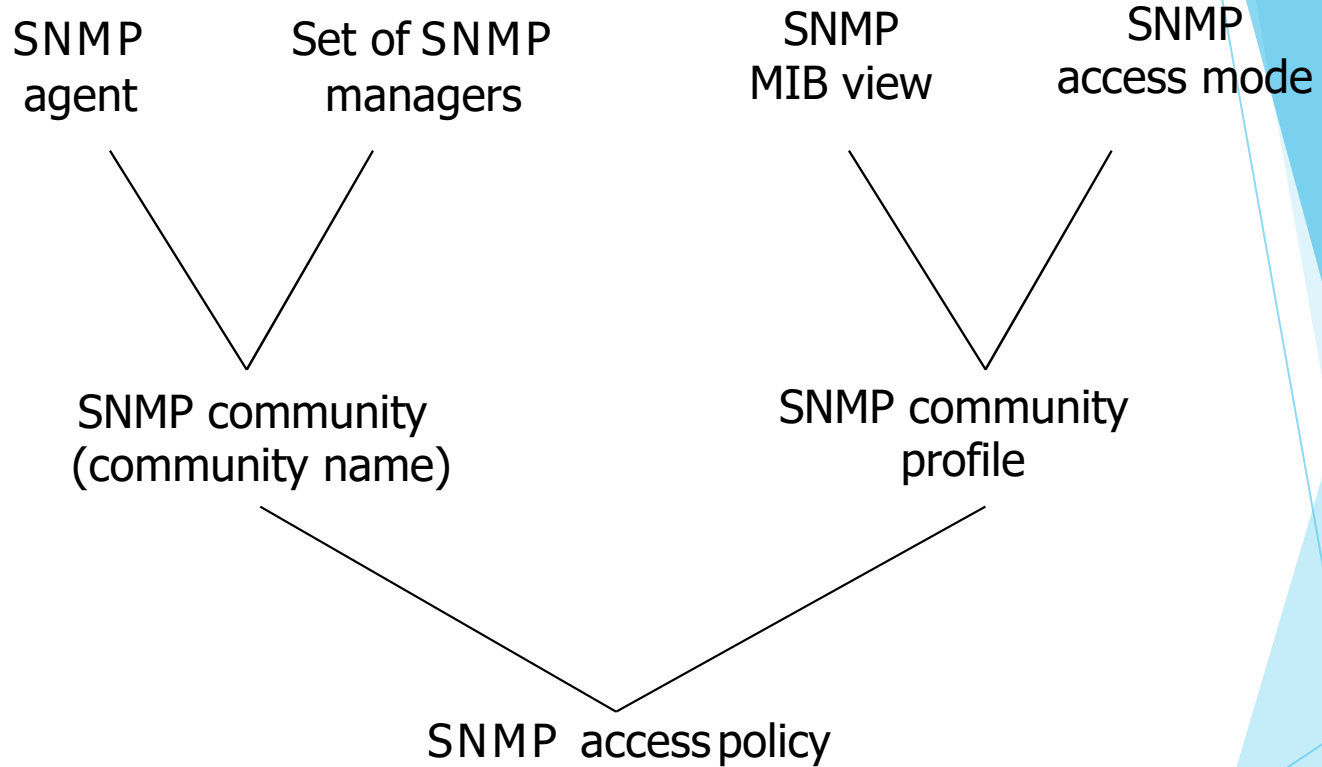
- SNMP MIB view

Subset of MOs on a given network element (made up of different MIBs and different subtrees).

The setting of MIB views must be explicitly supported in one component, otherwise all MOs in the MIB will be view.

- Access Mode

r-o or r-w



Example ("red", (MIB-view-System-2, r-o))

## Characterization of SNMP access protocol

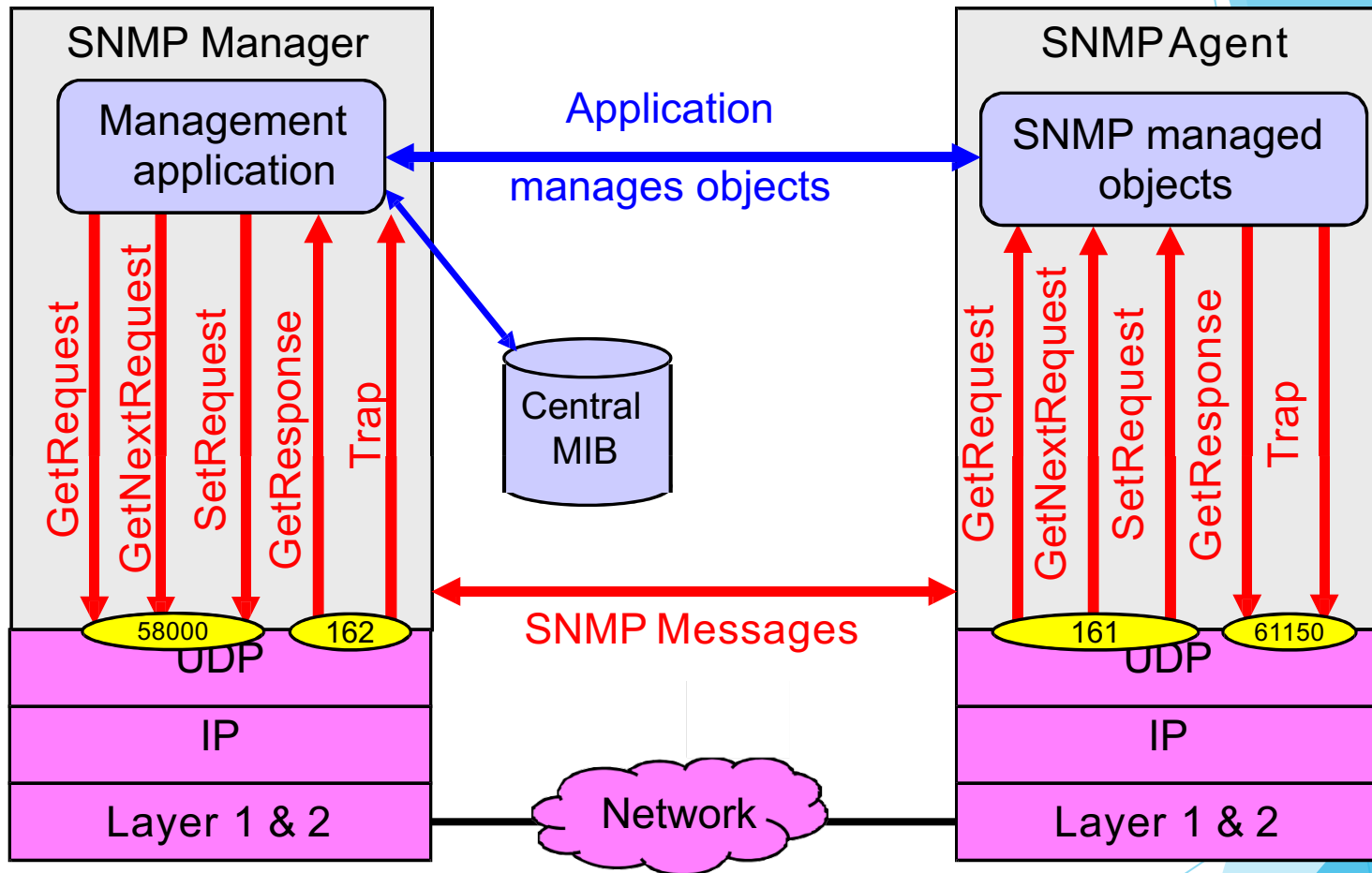
The SNMP protocol is a

- symmetrical

(Request / Response set up the same way → Manager und Agent PE gleich)

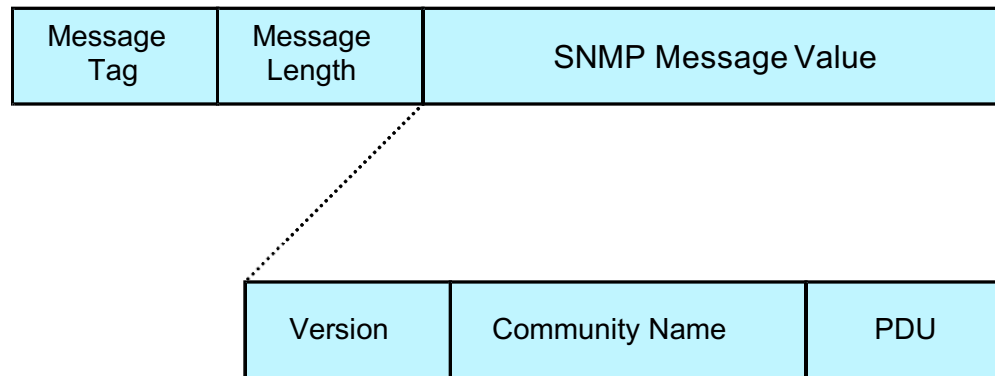
- asynchronous  
(possible to send at any time, no synchronization)

- request / response protocol.



According to Prof. John A. Copeland, Stallings, bzw. Dr. Andreas Steffen, 2000-2002 Zürcher Hochschule Winterthur, Kommunikationssysteme (KSy)

## Structure of the SNMPv2 message and ASN.1 definition



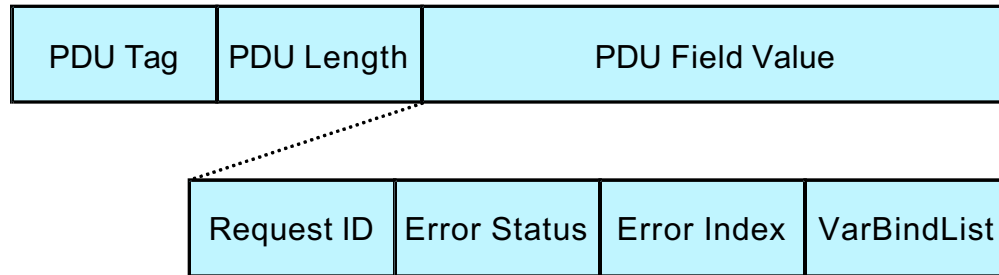
```
Message ::=
    SEQUENCE {
        version INTEGER {version(1)}
        community OCTET STRING
        data PDUs
    }
```



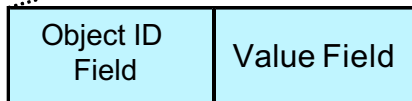
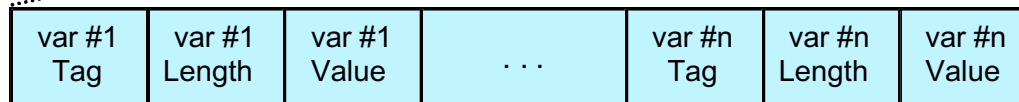
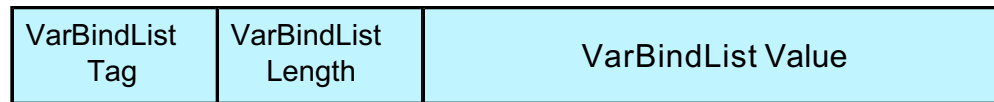
## ASN.1 description of the PDUs

```
PDUs ::= CHOICE {  
    get-request      GetRequest-PDU,  
    get-next-request GetNextRequest-PDU,  
    response         Response-PDU,  
    set-request      SetRequest-PDU,  
    .  
    .  
    snmpV2-trap      SNMPv2-Trap-PDU,  
    .  
}  
  
GetRequest-PDU      ::= [0] IMPLICIT PDU  
GetNextRequest-PDU  ::= [1] IMPLICIT PDU  
  
Response-PDU        ::= [2] IMPLICIT PDU  
  
SetRequest-PDU       ::= [3] IMPLICIT PDU  
    .  
    .  
SNMPv2-Trap-PDU     ::= [7] IMPLICIT PDU  
    .
```

## Structure of a PDU and its ASN.1 definition



```
PDU ::=
    SEQUENCE {
        request-id Integer32,
        error-status INTEGER {noError(0),
                               tooBig(1),
                               noSuchName(2),
                               badValue(3),
                               readOnly(4),
                               .
                               },
        error-index INTEGER (0..max-bindings),
        variable-bindings VarBindList
    }
```



```

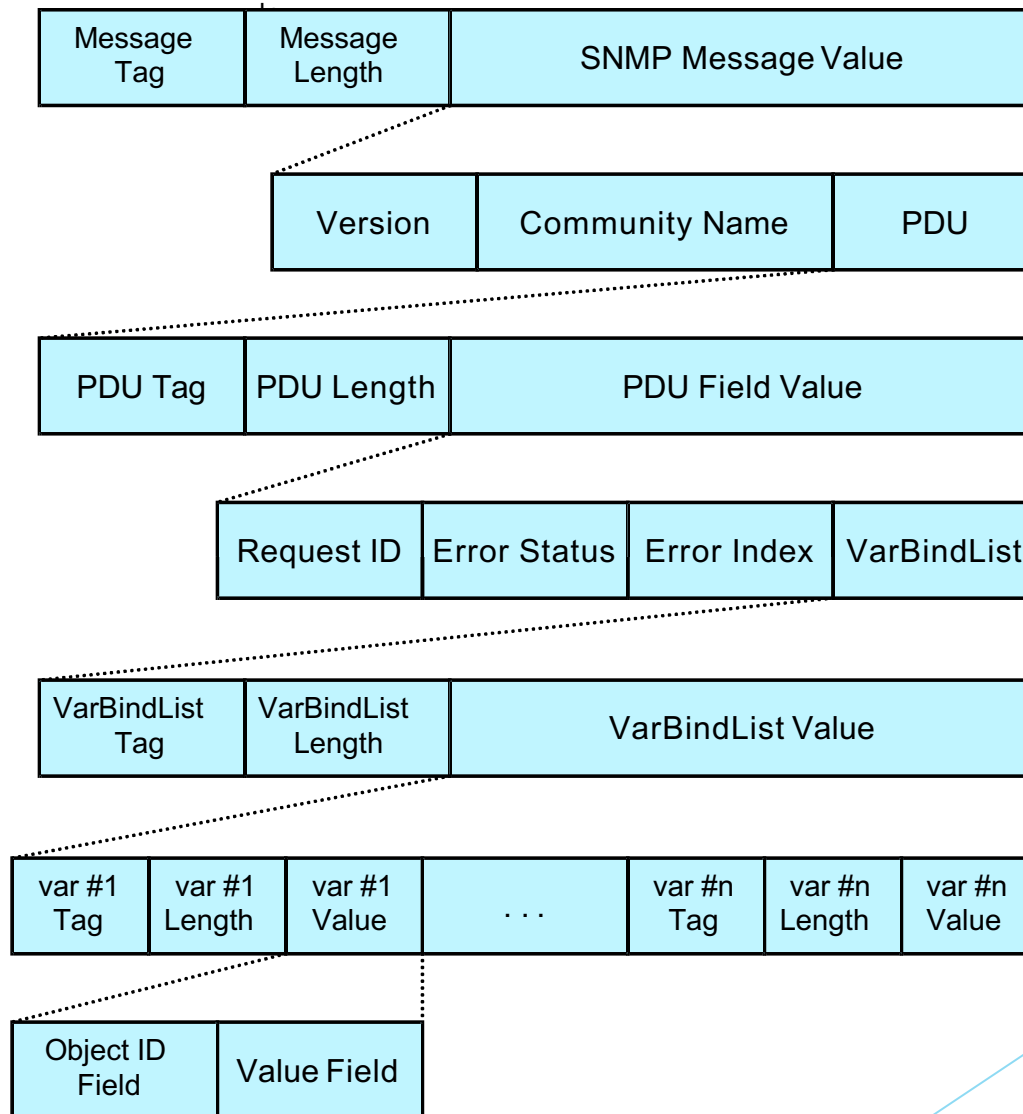
VarBind ::= SEQUENCE {
    name ObjectName
    value ObjectSyntax
}

```

```

VarBindList ::= SEQUENCE (SIZE (0..max-bindings)) OF
    VarBinding

```



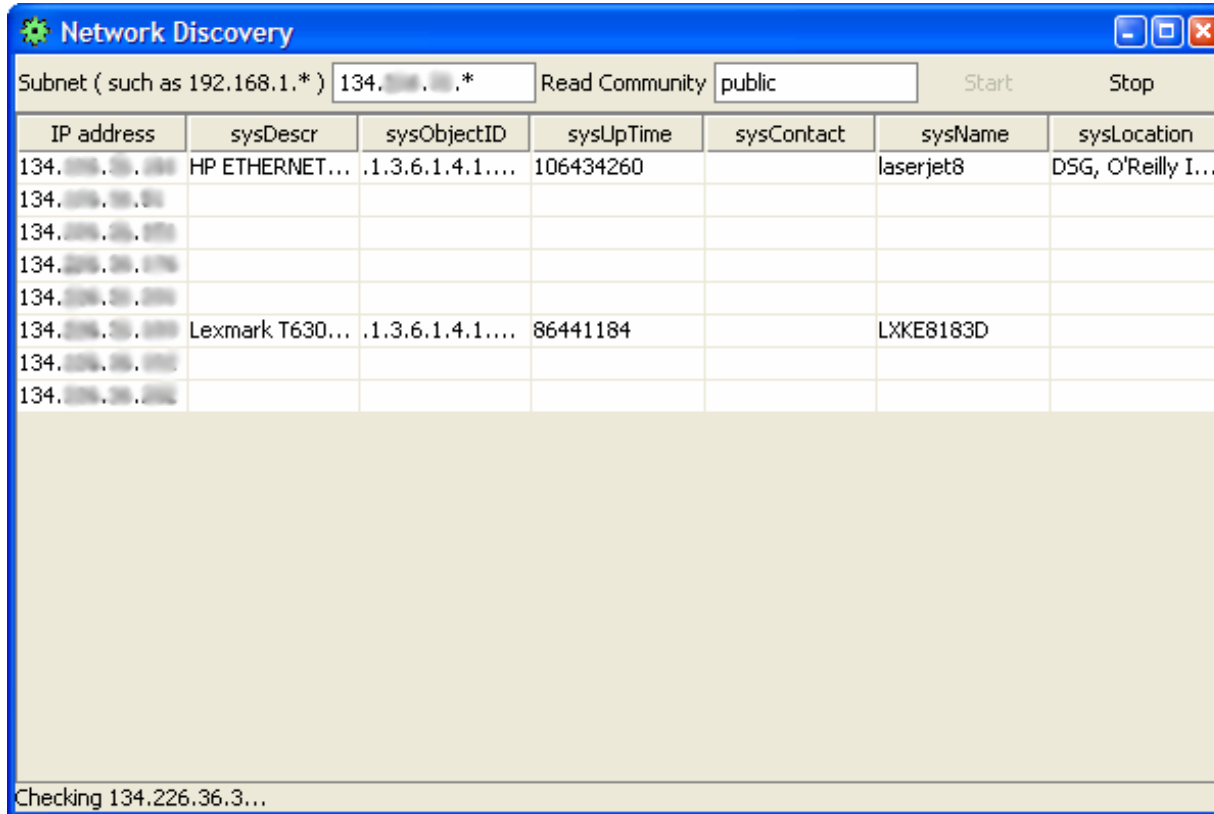
## Operation of the basic operations

- GetRequest
  - PDU-Tag=0
  - errorStaus=0
  - errorIndex=0
  - Value of variable=NULL
- GetResponse
  - PDU-Tag=1
  - Returns variable value to previous GetRequest, GetNextRequest or SetRequest in positive case,
  - Error message in the negative case
- GetNextRequest
  - PDU-Tag=2
  - Provides the following object instance in the OID tree lexicographically.
- SetRequest
  - PDU-Tag=3
  - Sets variables to specified values.
  - Error message (for example read-only) in the negative case.

# SNMP Tools

- ▶ Command Line Interface
  - ▶ e.g. 'snmpwalk'
- ▶ Graphical User Interface
  - ▶ e.g. iReasoning's MIB Browser
    - ▶ Via [www.ireasoning.com](http://www.ireasoning.com)

# SNMP - MIB Browser (1)



Discovery...

- Subnet: 134.XXX.XXX.\*

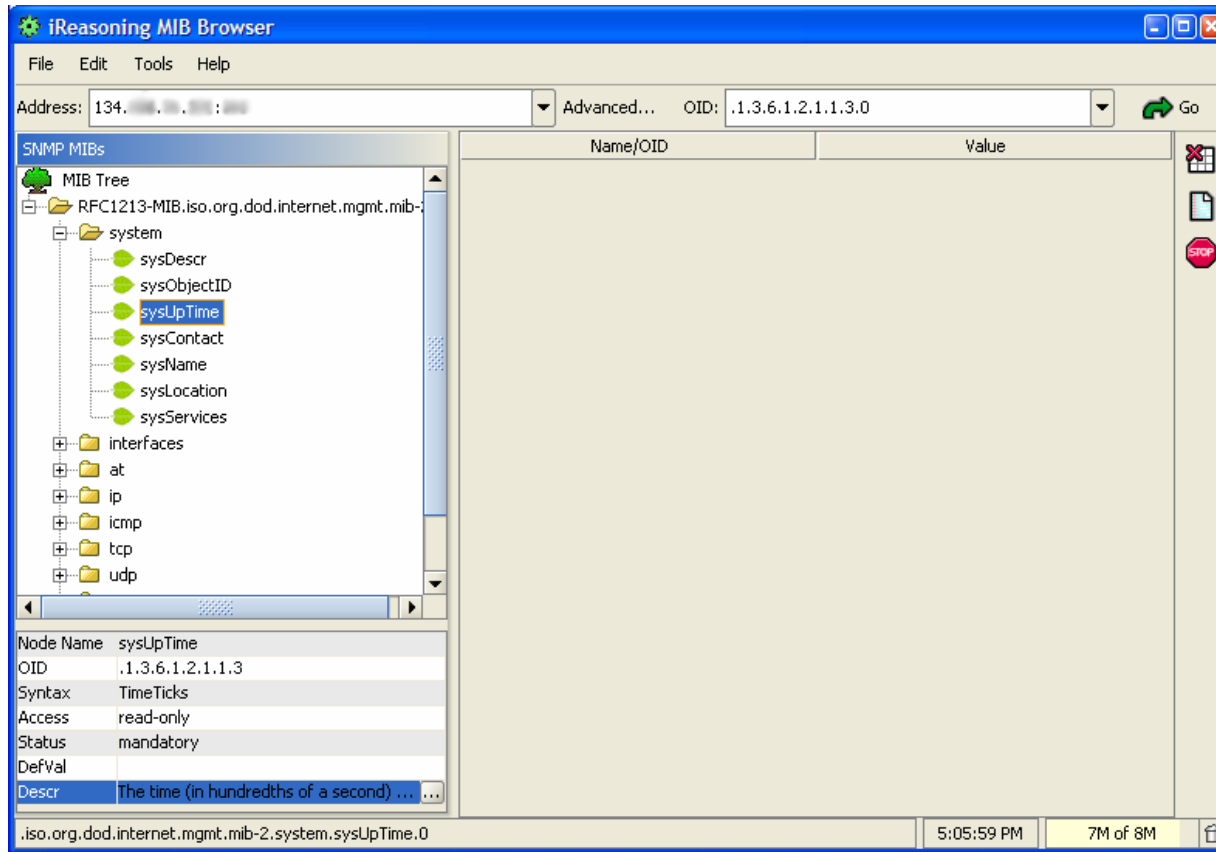
- Read Community: public

→ Start

Note IP Address.

← Stop

# SNMP - MIB Browser (2)

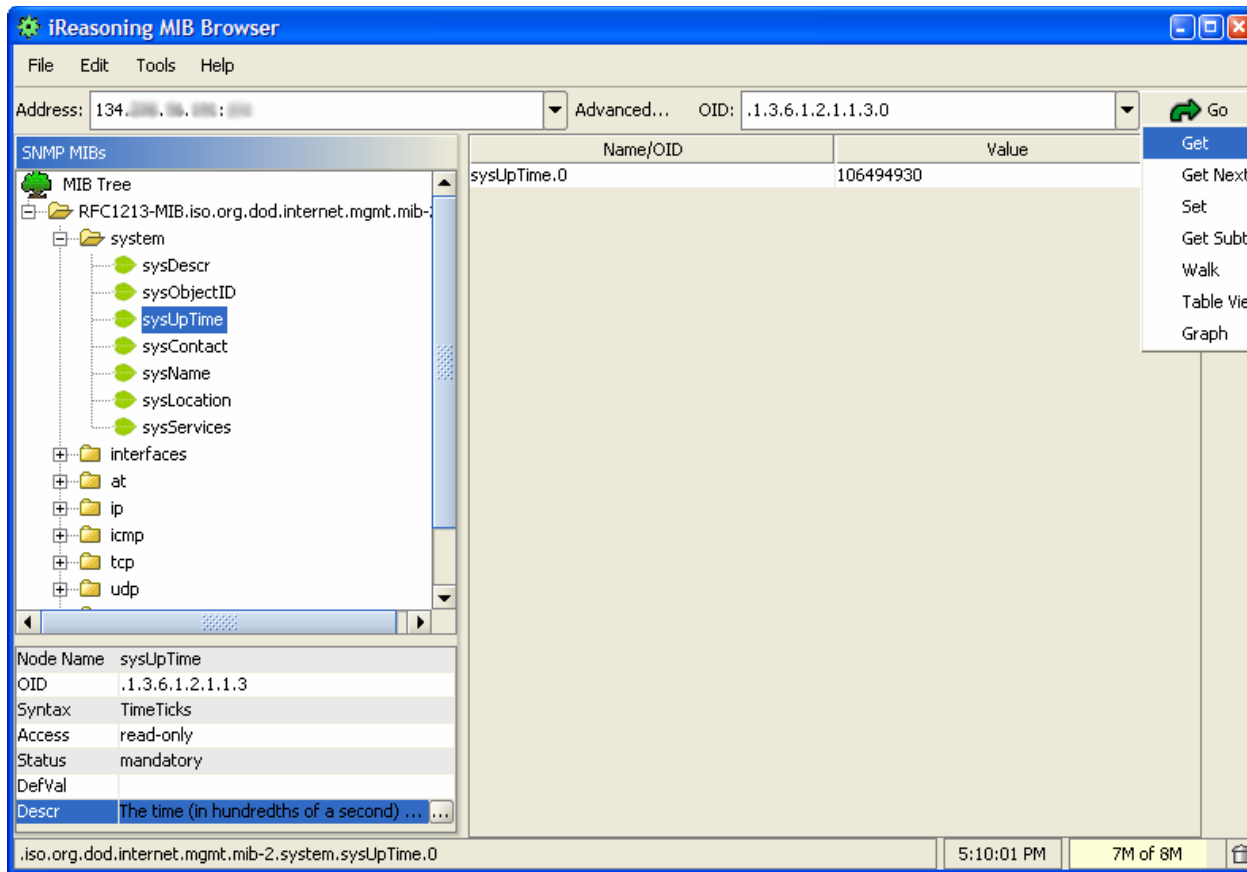


## Navigation...

- MIB Tree
  - System
    - sysUpTime
- Notice Lower LHS
- Notice OID



# SNMP - MIB Browser (3)



SNMP PDU's...

(1) Get

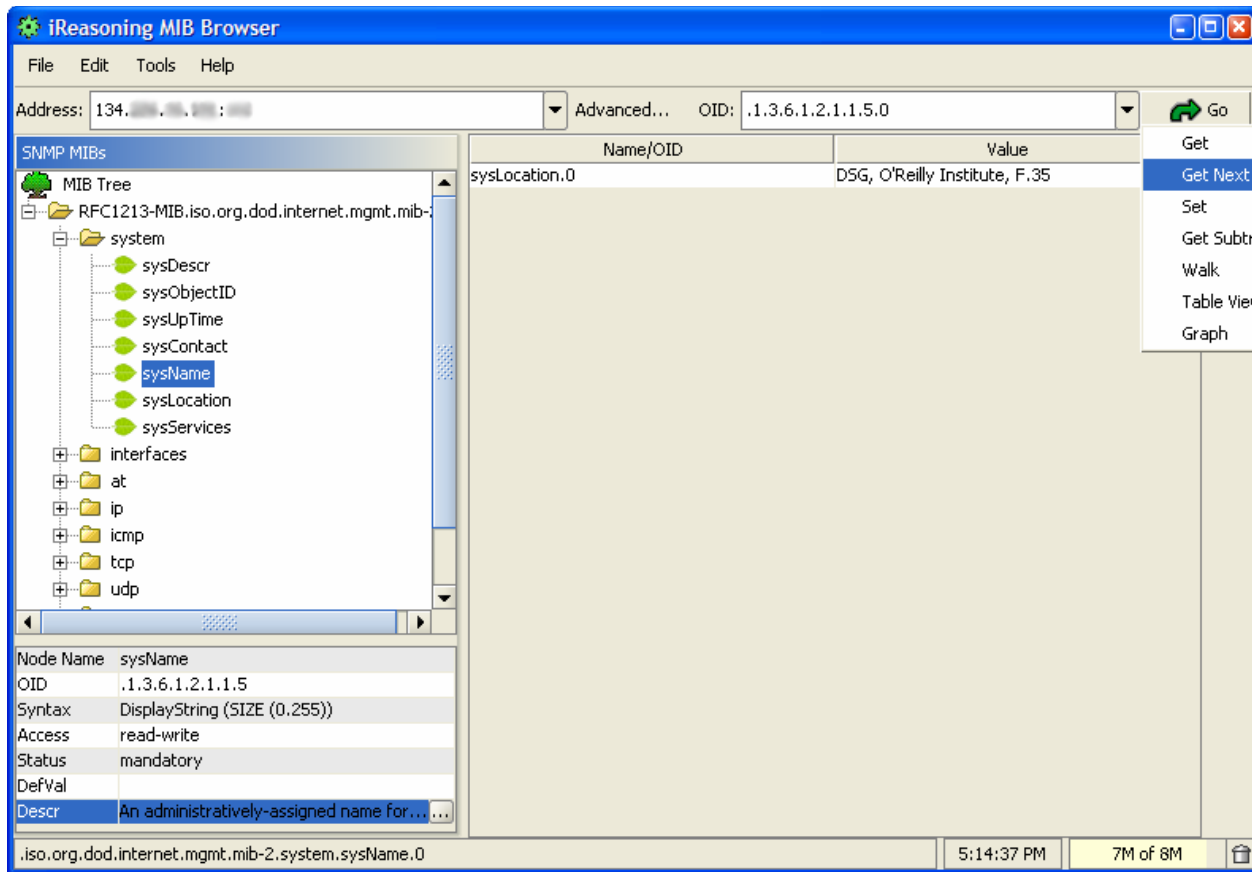
- Select 'Go'

→ 'Get'

- RHS has values.

- OID - Value

# SNMP - MIB Browser (4)



SNMP PDU's...

(2) GetNext

-Selected OID is:

.1.3.6.1.2.1.1.5

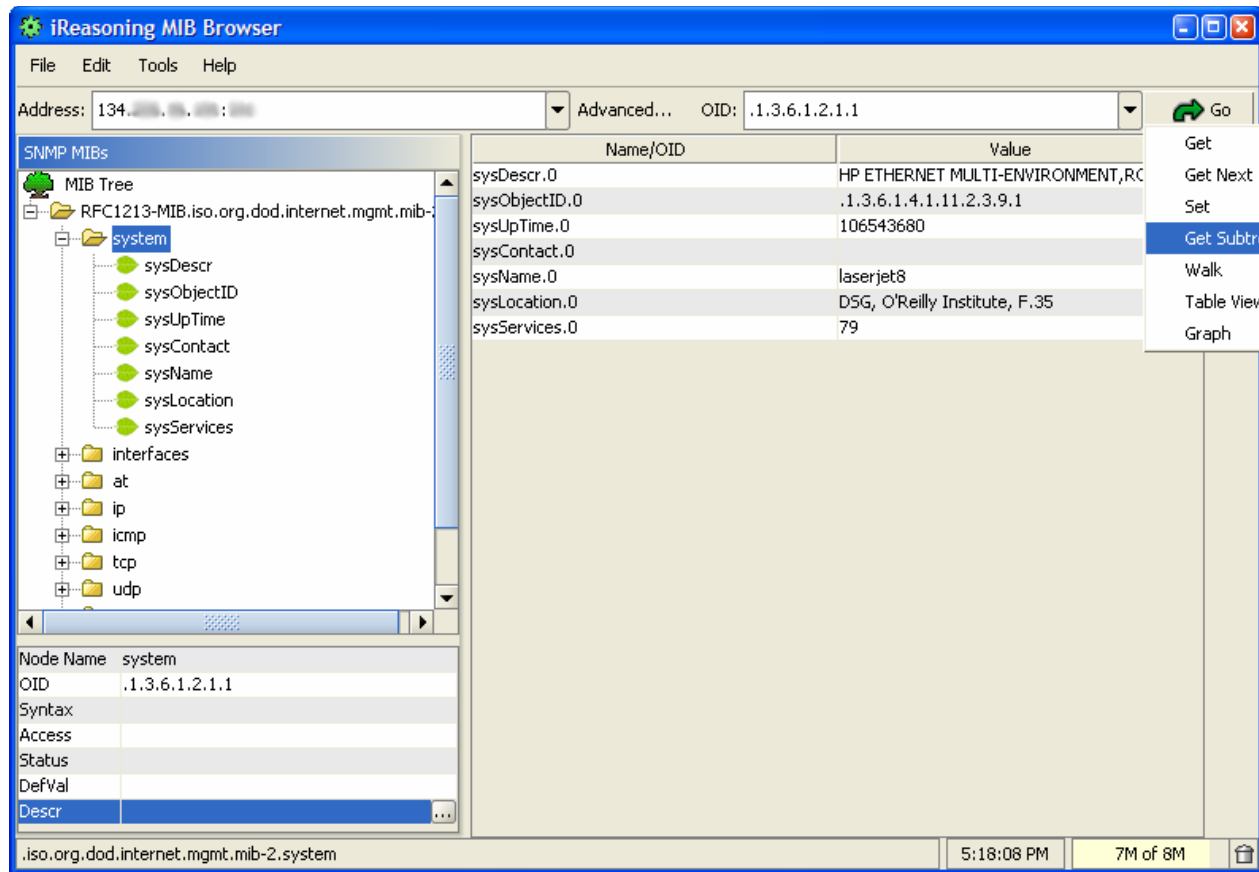
-Returned value:

(.1.3.6.1.2.1.1.6)

or

“DSG, O'Reilly Institute,  
F.35”

# SNMP - MIB Browser (5)



SNMP...

(3) Get SubTree

-Position of MIB:

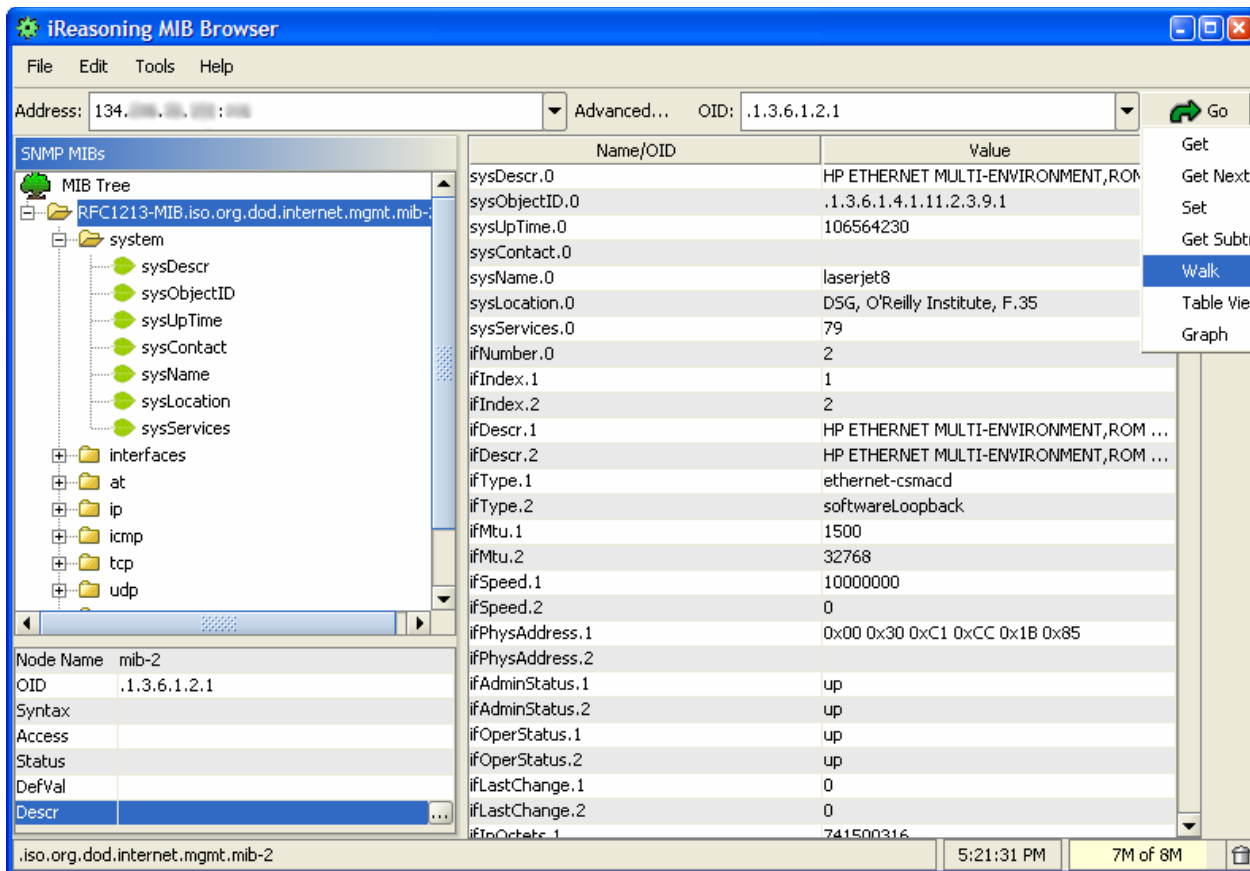
.1.3.6.1.2.1.1

(a.k.a. system)

-RHS values:

Returns all values  
below system.

# SNMP - MIB Browser (6)



SNMP...

(4) Walk

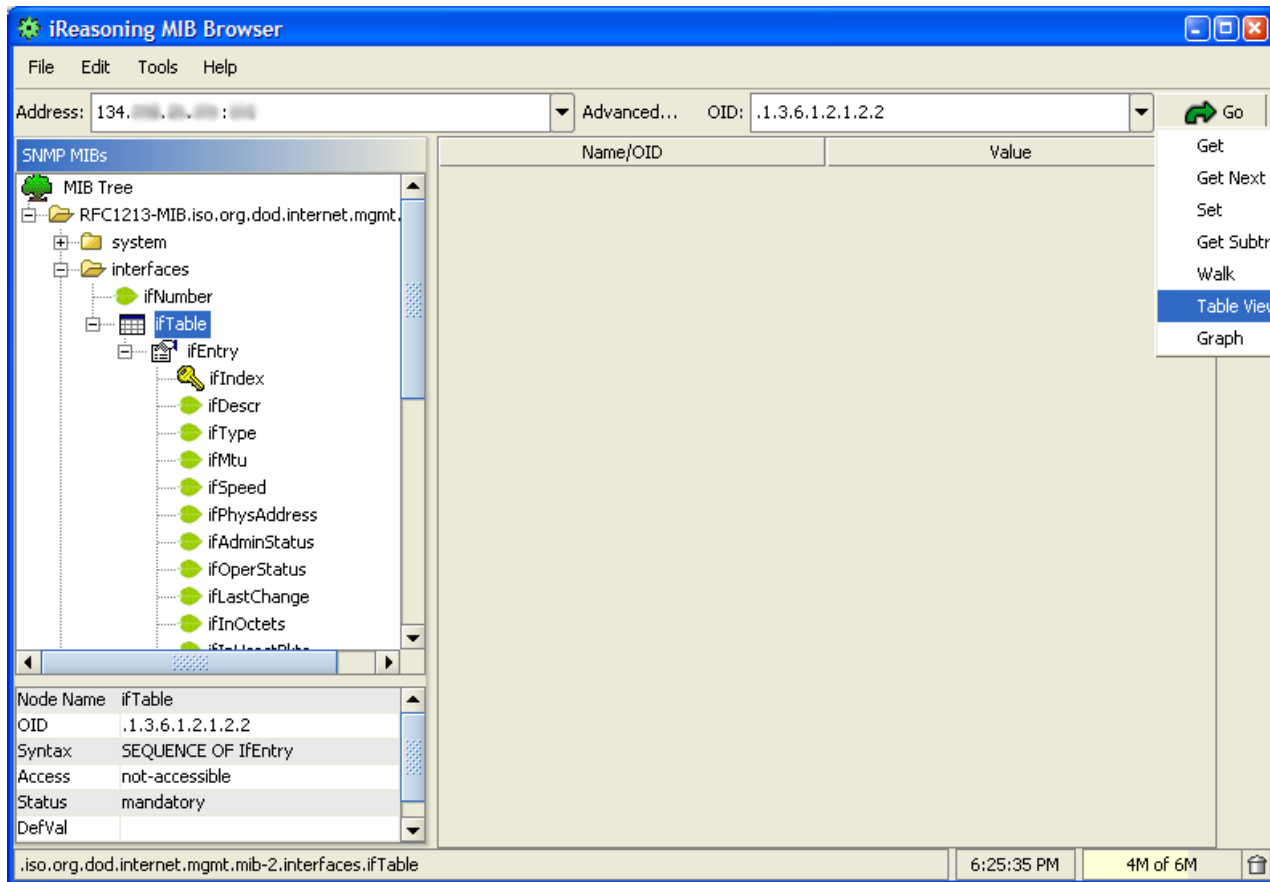
-MIB Location:

.1.3.6.1.2.1

(a.k.a. mib-2)

- Returns **\*ALL\***  
values under mib-  
2

# SNMP - MIB Browser (7)



Tables...

- MIB Location:  
.1.3.6.1.2.1.2.2  
(or interfaces)

- Select ifTable,  
→ Go, then Table View.

- Refresh/Poll

Thank you for your attention!